

Motif Binding (Structural Edition)

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Abstract

Motif Binding formalises how structural motifs attach to, detach from, and propagate through identity-bearing systems. It defines the structural grammar governing binding strength, binding stability, motif load, motif collapse, and motif interference. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.

1 Introduction

Motif Binding describes how motifs—structural units of identity—connect to the underlying system. A motif binds when its structural signature becomes stable relative to the system’s recurrence and boundary behaviour.

This edition defines:

- Motif Attachment
- Binding Strength
- Binding Stability
- Motif Load
- Binding Collapse
- Motif Interference
- Binding Drift

Motif Binding is domain-agnostic: it applies to reasoning engines, mathematical systems, physical models, and simulation architectures.

2 Motif Attachment

Let M denote motif structure and S system stability. Attachment occurs when:

$$S(M) > S_{\text{attach}}.$$

Binding radius B is defined structurally as:

$$B = f(S(M)),$$

where f is a monotonic structural mapping.

2.1 Structural Conditions

Motif attachment occurs when:

1. Recurrence stabilises.
2. Boundary motifs align.
3. Compression gradients flatten.

3 Binding Strength

Binding strength K is defined as:

$$K = \frac{dB}{dt}.$$

Strength increases when:

- Recurrence capacity rises
- Boundary stability improves
- Motif load decreases

4 Binding Stability

Binding stability requires:

$$K > K_{\min}.$$

Stability decreases when:

- Motif load increases
- Boundary instability rises
- Recurrence weakens

5 Motif Load

Motif load L is defined structurally as:

$$L = g(M, S).$$

Load increases when:

- Motif complexity increases
- Boundary motifs interfere
- Recurrence capacity decreases

6 Binding Collapse

Collapse condition:

$$\Delta B < 0.$$

Collapse velocity:

$$V_c = -\frac{dB}{dt}.$$

Collapse is triggered by:

- Excessive motif load
- Boundary instability
- Recurrence failure

7 Motif Interference

Interference strength I :

$$I = h(M_1, M_2, d),$$

where d is separation distance.

Interference includes:

- Motif merging
- Motif repulsion
- Motif resonance

8 Binding Drift

Drift velocity:

$$V_d = \frac{dB}{dt}.$$

Drift occurs when:

- Information density changes
- Recurrence capacity evolves
- Boundary motifs reconfigure

9 Conclusion

Motif Binding provides the structural grammar governing motif attachment, stability, collapse, drift, and interference. This edition presents the basis-independent formulation suitable for academic publication and cross-domain application.